

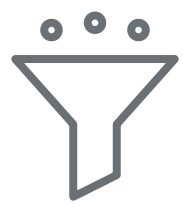
50 STATISTICAL BIASES TO BE AWARE OF

SO YOUR DATA TELLS THE TRUTH, NOT WHAT YOU HOPED

Sampling

Selection Bias

The group you study differs systematically from the group you want to draw conclusions about.

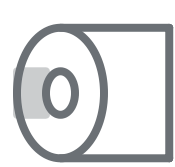


"The new coating works great!" (It was only trialled on day shift, with the best crew.)

Measurement

Sampling Bias

The way samples are collected makes some units more likely to be picked than others.



"All our coupons come from the head of the coil. It's the easiest end to cut."

Modelling

Convenience Sampling

We sample whatever is easiest to reach instead of what represents the population.



"We surveyed everyone in the break room." Night shift was asleep.

Testing

Undercoverage Bias

Part of the population has little or no chance of ending up in the sample.



"Our phone survey only called landlines, so almost no one under 40 answered."

Time

Reporting

Volunteer Bias (aka Self-Selection)

People who choose to take part differ from those who don't.



"The optional SPC course had great exam scores." Only the keenest operators signed up.

Nonresponse Bias

The people who don't respond differ from the people who do.



"Only 12% replied, mostly customers with a complaint. Satisfaction must be terrible!"

Survivorship Bias

We study only what survived a process and overlook what failed.



"Armor the bullet holes on returning bombers?" Planes hit elsewhere never came back.

Berkson's Bias (aka Admission Rate Bias)

Studying only units that entered a system, like a hospital, creates links that don't exist in the wider population.



"In hospital, two unrelated diseases look connected." Having either one gets you admitted.

Healthy Worker Effect

People who are working are healthier than the general public, so comparing to the public understates risk.



"Mill workers outlive the general population, so the job must be harmless."

Spectrum Bias

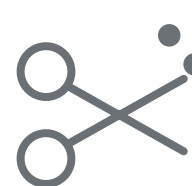
A test is evaluated only on clear-cut cases, so it looks more accurate than it will in practice.



"The UT check caught 100% of defects!" It was only tested on obvious cracks.

Exclusion Bias

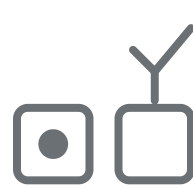
Removing data points or groups in a way that skews the results.



"Delete the soft readings as outliers and Cpk jumps from 0.9 to 1.6."

Allocation Bias

Groups are assigned non-randomly, so they differ before the treatment even starts.



"The supervisor picked which heats got the new practice." He picked the easy grades.

Measurement Bias

The instrument or method reads consistently high or low.



"The tensile frame reads 2% high, so every heat tested on it looks stronger."

Observer Bias

The person recording results sees what they expect to see.



"The inspector was told this heat was suspect, and found twice the surface marks."

Recall Bias

People remember past events inaccurately, colored by what happened afterwards.



"After the pipe failed, everyone remembered the line running rough all week."

Misclassification Bias

Items are sorted into the wrong category, distorting counts and associations.



"Half the logged laps were actually seams." The Pareto chart points the wrong way.

Social Desirability Bias

Respondents give answers that make them look good rather than true ones.



"Everyone says they always wear hearing protection." The audiograms disagree.

End-Digit Preference

Readers round values to favorite digits like 0 and 5.



"Strangely, 60% of manual temperature readings end in zero."

Hawthorne Effect*

People change their behavior because they know they are being observed.



"Output jumped during the time study, then dropped the moment the clipboard left."

Detection Bias

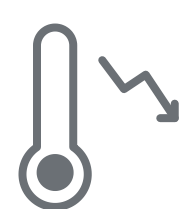
One group is examined more closely than another, so more is found there.



"New-caster heats get 100% UT, old-caster heats get spot checks. Guess which has more defects."

Calibration Drift

An instrument slowly wanders off calibration, adding a bias that grows over time.



"The thermocouple reads a little lower every month." The trend chart shows the process cooling.

Ceiling & Floor Effects

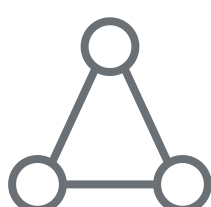
The scale can't record values beyond its limits, hiding real differences.



"Every sample maxed out the 300 J pendulum, so the new practice had no effect."

Omitted Variable Bias (aka Confounding)

A factor that drives both the cause and the outcome is left out, so its effect is credited elsewhere.



"Ice cream sales and drownings rise together." Summer heat drives both.

Confounding by Indication

The reason a treatment was chosen is itself linked to the outcome.



"Heats given extra calcium have more inclusions." We only add it to the dirtiest heats.

Collider Bias

Selecting on, or adjusting for, a shared effect of two variables creates a false link between them.



"Among movie stars, talent and looks seem opposed." You need at least one to get famous.

Simpson's Paradox*

A trend in every subgroup reverses when the groups are combined.



"Line A wins on every grade but loses overall." It runs far more of the hardest grade.

Regression Dilution (aka Attenuation)

Random error in a predictor flattens the fitted slope toward zero.



"Our noisy thermocouple makes temperature look like it barely affects hardness."

Regression to the Mean*

Extreme results tend to be followed by more typical ones, which can look like an effect.



"We retrained the worst crew after a terrible month, and scrap improved!" It would have anyway.

Ecological Fallacy

Assuming a relationship seen between group averages also holds for individuals.



"Plants with more automation have fewer injuries, so every automated job is safer."

Overfitting

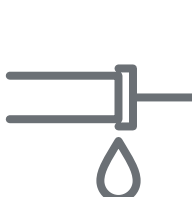
A model learns the noise in its training data and fails on new data.



"R² of 0.99 on last year's heats." Useless on this month's.

Data Leakage

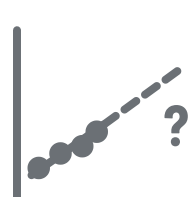
Information that won't exist at prediction time sneaks into model training.



"The scrap model used the rework code as an input." That code is assigned after scrapping.

Extrapolation Bias

Applying a model outside the range of data it was built on.



"The model fits 0.08 to 0.12% carbon, so let's predict 0.25%."

Small-Sample Estimator Bias

Some estimators are systematically off when samples are small.

$$n = 3$$

"Divide by n instead of $n - 1$ and five samples make the variance look too small."

Data Dredging (aka Multiple Comparisons)

Testing many variables until something turns up significant.



"We tested 40 variables and 2 were significant!" At $\alpha = 0.05$, chance alone expects 2.

P-Hacking

Tweaking the analysis until p drops below 0.05.

$$p = .049$$

"Drop three points, add a covariate, try a log... $p = 0.049$. Ship it!"

HARKing

Hypothesizing After the Results are Known, then presenting it as the plan all along.



"We always suspected a Tuesday effect," said right after it appeared in the data.

Early Stopping Bias

Stopping a study as soon as results look good overstates the effect.



"It looked great after 3 heats, so we called it." The next 20 would have disagreed.

Winner's Curse

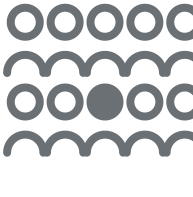
The first significant estimate of an effect usually overstates its true size.



"The pilot showed a 30% yield gain." The full rollout delivered 8%.

Base Rate Fallacy

Ignoring how common something is when interpreting a test result.



"The detector is 95% accurate, so a flagged pipe is 95% likely bad." Not if only 1 in 1,000 is.

Underpowered Study (aka Wrong Sample Size)

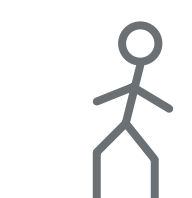
A sample too small to detect a real effect, or one that only catches exaggerated ones.



"Three samples showed no difference, so the new practice doesn't work."

Attrition Bias

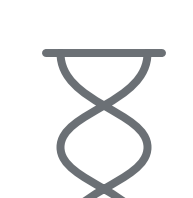
People or units that drop out of a study differ from those who stay.



"Only people who stuck with the diet were weighed at the end." It looks miraculous.

Immortal Time Bias

A period when the outcome couldn't happen is wrongly credited to the treatment group.



"Award winners live longer!" You have to stay alive long enough to win one.

Lead Time Bias

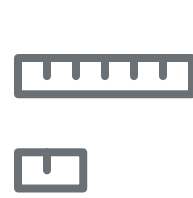
Earlier detection makes survival look longer even when the outcome doesn't change.



"Screening found it two years sooner, so survival rose two years." The end date never moved.

Length Time Bias

Periodic checks catch slow-developing cases more easily, making outcomes look better.



"Inspections catch the slow-growing cracks." The fast ones fail between inspections.

Look-Ahead Bias

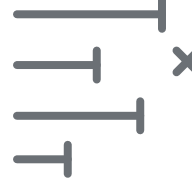
Using information that wouldn't have been known at the time of the decision.



"The backtest bought stocks using earnings reports released weeks later."

Informative Censoring

Units leave follow-up for reasons tied to the outcome, but the analysis treats it as random.



"Pipes pulled from the rig early were dropped from the analysis." They were the ones starting to leak.

Publication Bias (aka File Drawer Effect)

Studies with positive or exciting results are more likely to be published.



"Twenty labs tested the idea." The one lucky positive result is the one you read.

Outcome Reporting Bias

Only the outcomes that came out well get reported.



"We measured yield, scrap, cost and delays." The report mentions yield.

Spin Bias

Presenting results as more favorable than the data support.



"Not statistically significant, but a strong trend toward improvement!"

Sponsorship Bias (aka Funding Bias)

Results lean toward the interests of whoever paid for the study.



"The supplier-funded trial found their refractory lasts longest."

*Technically an effect or statistical paradox rather than a bias, but it produces biased conclusions.

Sources: Oxford CEBM Catalogue of Bias (catalogofbias.org); Sackett DL, "Bias in Analytic Research," J Chronic Dis 1979; literature on regression dilution and multiple testing. Examples are illustrative.